

Foundations of Modern Vision Models: An Empirical Study of Residual Networks, Vision Transformers, CLIP, and Variational Autoencoders

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PA 0

Github Repo: <https://github.com/ahmadhassann96/ATML-PA0>

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Abstract

Four empirical studies spanning discriminative and generative vision modelling. (i) A linear head on a frozen ImageNet-pretrained ResNet-152 reaches 84.9% on CIFAR-10 while training only 20,490 of 58.2M parameters; removing residual connections from the frozen backbone reduces accuracy to 13.5%, and layer-wise t-SNE shows that class structure emerges mainly between layer 3 and layer 4. (ii) For ViT-B/16, raw final-layer CLS attention is not well aligned with objects. It focuses on fixed, image-independent background positions while attention rollout recovers clear object silhouettes. Head diversity is highest in early layers and mostly disappears by layer 11. Central patch masking is much more damaging than random masking (58.6% vs. 78.2% at 50%), while border masking slightly improves accuracy. (iii) CLIP ViT-B/32 reaches 97.31% zero-shot top-1 accuracy on the full STL-10 test set using “a photo of a { }” prompts; longer descriptive prompts reduce accuracy to 92.76%. The modality gap (centroid cosine 0.269) is unchanged by L2 normalisation. An orthogonal Procrustes rotation fitted on 5,000 labelled pairs closes the gap (cosine 0.998) but improves accuracy by only +0.51 points. (iv) An MLP VAE on MNIST reproduces Doersch’s finding that the ELBO is relatively insensitive to latent dimensionality over a broad range: test negative ELBO changes by 15.2 nats between $d = 5$ and $d = 20$, but by 40.8 nats between $d = 2$ and $d = 20$. Negative and anomalous results (including a diverged run and a metric that failed to discriminate) are also explicitly reported.

1 Introduction and Experimental Setup

The four tasks trace an arc through modern vision: the convolutional inductive bias (ResNet), its replacement by global attention (ViT), language as an open-vocabulary supervision signal (CLIP), and explicit latent-variable generative modelling (VAE). This report documents what was run, what was measured, and where the measurements failed to support clean conclusions.

Compute environment. All experiments ran on Google Colab (T4 GPU). Software: PyTorch 2.11.0+cu128, torchvision 0.26.0, scikit-learn 1.6.1, SciPy 1.16.3, umap-learn 0.5.12, HuggingFace Transformers, OpenAI CLIP. Seeds were fixed (0 for Tasks 1.4, 2 and 4; 42 for the t-SNE/UMAP projections in Task 1). Free-tier limits forced several experiments to be restructured, most consequentially Task 1.4 (Section 2.4), which changes constant factors but not what is compared; where a shortcut does compromise comparability it is stated explicitly (Section 6).

2 Task 1: Inner Workings of ResNet-152

2.1 Baseline: Frozen Backbone, Trained Head

A torchvision ResNet-152 with IMAGENET1K_V2 weights was loaded, every backbone parameter was frozen (`requires_grad = False`), and the final fully-connected layer was replaced by a fresh $2048 \rightarrow 10$ linear map: 20,490 trainable parameters out of 58,164,298, or 0.035% of the network. CIFAR-10 images were resized to 224×224 and normalised with ImageNet statistics; Adam ($\text{lr} = 10^{-3}$), batch size 64, cross-entropy, 5 epochs on the full 50K training split with the 10K test split as validation.

Validation accuracy reached $\approx 82.4\%$ after one epoch and settled at $\approx 84.9\%$ by epoch 5 (Figure 1). Train and validation curves separate only slightly, as expected with 20K trainable parameters.

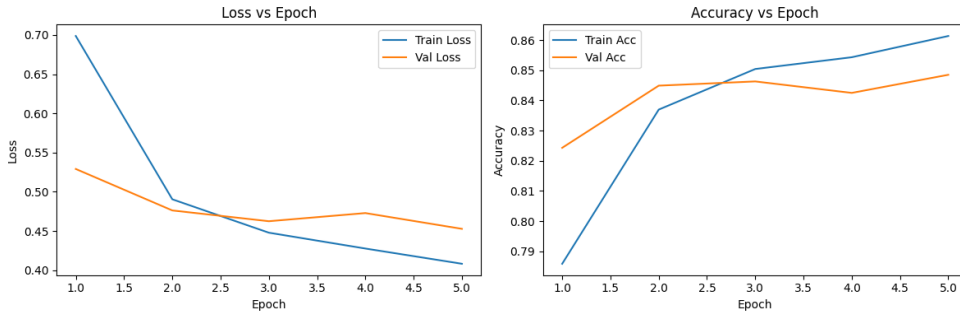


Figure 1: Task 1.1 baseline. Validation accuracy saturated near 84.9% within three epochs while training accuracy keeps climbing, showing that the representation of the model is fixed and its only remaining capacity is a linear decision boundary.

Table 1: Task 1.2. Baseline vs. residual-ablated backbone, CIFAR-10, five epochs, head-only training.

Configuration	Train Acc.	Val. Acc.	Val. Loss	Time/epoch
Baseline (skips intact)	0.861	0.849	0.452	—
4 blocks, skips removed	0.133	0.135	2.379	435 s

Why not train ResNet-152 from scratch? There are two reasons. One is complexity: 58M parameters against 50K images at 32×32 is an enormously over-parameterised hypothesis class, and without the ImageNet prior the optimiser has no reason to prefer a generalising solution. Our own measurement confirms this, as training from scratch reaches 42.3% under a matched budget (Section 2.4) against 93.3% for a linear probe. The second reason is computational expense: one epoch over 50K images at 224×224 costs ≈ 565 s on a T4, so the 90+ epochs needed to be competitive would take ≈ 14 GPU-hours per configuration, which is expensive and unnecessary when the frozen-feature route reaches higher accuracy in minutes.

What freezing tells us about transferability? It tells us that a *linear* map on frozen ImageNet features achieves 84.9% on a disjoint 10-class problem, which is a substantive finding: the ImageNet representation has, without ever seeing CIFAR-10, arranged these ten categories into near-linearly-separable regions. The features are general-purpose visual descriptors rather than ImageNet-specific detectors. This implies that the backbone learned *what visual properties matter*, and the only dataset-specific knowledge required is *how to weight them*, which is exactly what the 20K-parameter head supplies.

2.2 Residual Connections in Practice

Skip connections were disabled in four bottleneck blocks (`layer2[0]`, `layer2[1]`, `layer3[0]`, `layer3[1]`) by rebinding each block’s `forward` to a variant omitting the `out += identity` step, leaving the convolution–BN–ReLU stack intact. All other settings match the baseline.

The ablated network sits at chance (10%) plus a few points and its loss never leaves the neighbourhood of $\ln 10 = 2.303$ (Table 1), meaning the loss does not even converge.

Interpreting this correctly. It is tempting to read this as demonstrating that skip connections are needed for gradient flow. That reading is not supported by *this* experiment. Because the backbone is frozen, no gradients flow through the modified blocks at all; the only trainable parameters are in the head, downstream of every ablation. What is actually measured is the effect of residual connections on the *forward* pass of an already-trained network.

That effect is severe, and is explained by weight–architecture mismatch. Every pretrained weight was optimised assuming block i computes $\mathcal{F}_i(x) + x$, so the residual branch learned to output a small *correction* rather than a self-sufficient representation. Deleting the identity path leaves the correction alone, and with four blocks ablated the corruption compounds multiplicatively through the remaining depth; by `layer4` the activations bear no useful relation to the input. The gradient-flow argument is real, but it concerns *trainability from scratch*: the identity path makes $\partial\mathcal{L}/\partial x = \partial\mathcal{L}/\partial y \cdot (I + \partial\mathcal{F}/\partial x)$, guaranteeing a direct route for gradient signal to early layers. Testing it would require training both architectures from random initialisation, which was outside the compute budget, therefore it is necessary to flag this as the correct follow-up rather than claiming tha the present result speaks to it.

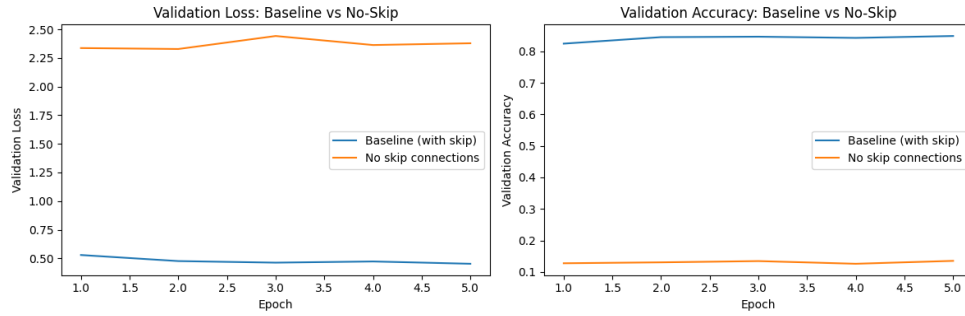


Figure 2: Task 1.2. Training dynamics with residual connections removed from four mid-depth blocks of the frozen backbone. Validation accuracy never exceeds 13.5% and the loss stays pinned near $\ln 10 \approx 2.30$; compare Figure 1.

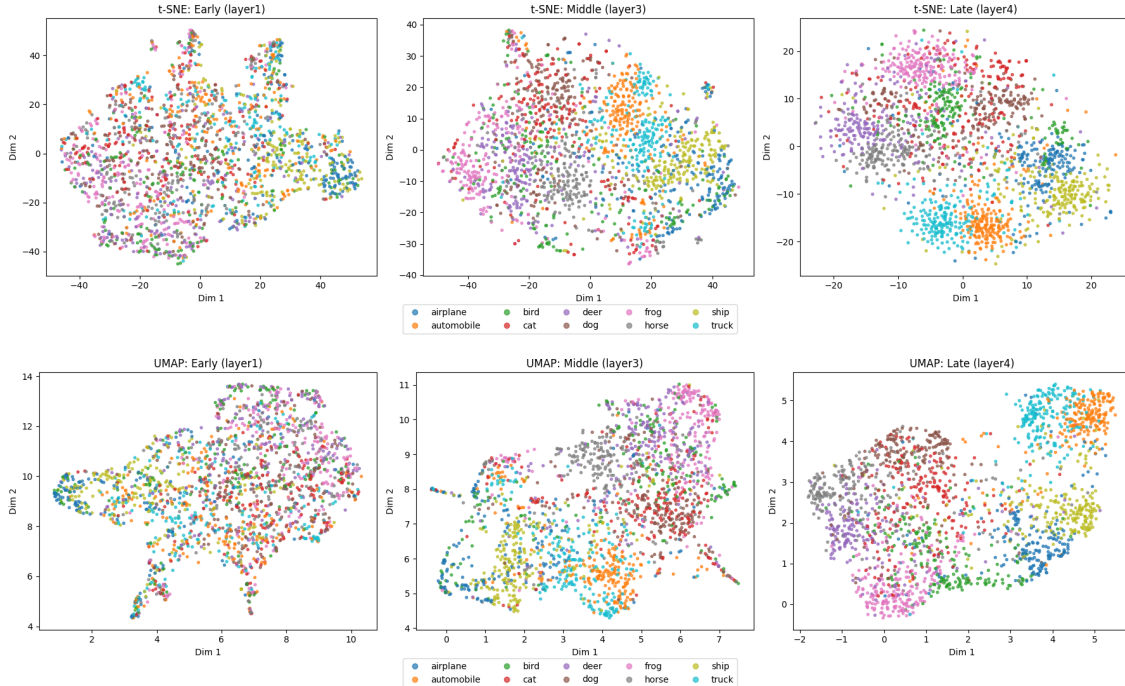


Figure 3: Task 1.3/1.5a. t-SNE (top) and UMAP (bottom) of pooled features from early, middle and late stages of the frozen ResNet-152, coloured by CIFAR-10 class. Class structure is essentially absent at `layer1`, partially formed at `layer3` and clearly organised at `layer4`. Both methods agree on this ordering.

2.3 Feature Hierarchies and Representations

Forward hooks were registered on `layer1` (early), `layer3` (middle) and `layer4` (late). Each feature map was globally average-pooled (256-d, 1024-d, 2048-d) over 2,048 validation images, then projected to two dimensions with t-SNE (perplexity 30, PCA init) and UMAP.

How separability evolves. The progression in Figure 3 is unambiguous. At `layer1` the ten colours are thoroughly interleaved; the visible structure corresponds to low-level image statistics such as brightness, dominant colour, and texture density which cut across semantic categories, so a red car and a red bird land near each other. At `layer3` coherent regions appear, with the vehicle classes in one broad territory and the animals in another, with substantial boundary overlap. At `layer4` the ten classes form identifiable clusters, with residual mixing concentrated exactly where one would predict semantically, e.g. cat with dog, deer with horse, or automobile with truck. This is the standard picture of a progressive hierarchical representation, with early layers encoding *appearance*, and late layers encoding *category*. The corollary explains Task 1.1—if `layer4` features are already this well organised, a linear head suffices.

A metric that did not work. Silhouette scores (Table 3) are near zero everywhere and negative for two layers, naively suggesting no clustering at any depth, and flatly contradicting both the visualisations and the 84.9% probe accuracy. This is because silhouette compares each point’s mean distance to its own

Table 2: Task 1.4. Transfer-learning comparison on CIFAR-10 (10,000 train / 2,000 validation, 160 px, 5 epochs). Learning rate 10^{-3} for head-only runs and 10^{-4} for runs that update backbone weights. [†]This run diverged; see discussion.

Configuration	Trainable params	Train Acc.	Val. Acc.	Val. Loss
Pretrained + head only	20,490	0.9775	0.9330	0.2103
Pretrained + final block	14,985,226	0.9269	0.9210	0.2526
Random init + full backbone	58,164,298	0.4538	0.4225	1.5802
Random init + head only [†]	20,490	0.1000	0.1000	NaN

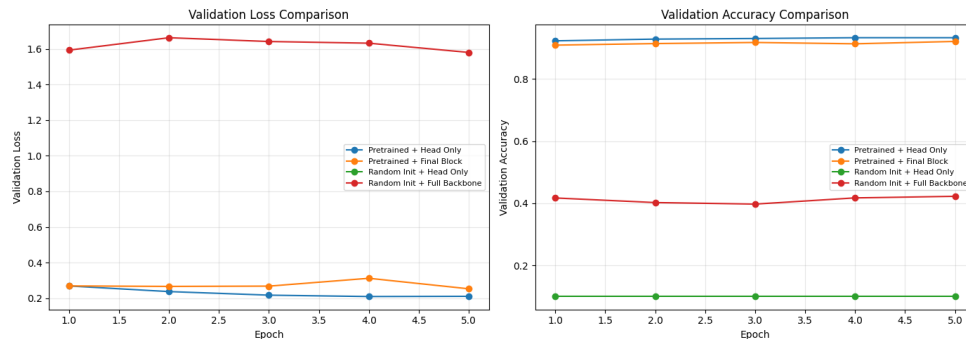


Figure 4: Task 1.4. Validation loss and accuracy across the four transfer configurations of Table 2. The two pretrained variants sit within roughly one accuracy point of each other; the from-scratch backbone is some 50 points behind after an equal number of epochs.

class against that to the nearest other class using Euclidean distance, and in 2048 dimensions Euclidean distances concentrate: the ratio between nearest and farthest neighbour tends to 1, so numerator and denominator become nearly equal regardless of class structure. Silhouette also measures *compact*, *isotropic*, *well-separated* clusters, whereas linear separability requires only that classes fall on opposite sides of a hyperplane, but classes can be elongated and interleaved along a manifold and still be perfectly separable. Per-layer k -NN or linear-probe accuracy would have been a better evaluation metric.

2.4 Transfer Learning and Generalization

Experimental restructuring. A direct implementation re-runs a full ResNet-152 forward pass over all CIFAR-10 images for every epoch of every configuration: at ≈ 565 s/epoch that is ≈ 3.5 GPU-hours for four configurations. Three changes brought it to ≈ 15 minutes: (i) **Feature caching for frozen-backbone runs**—a frozen backbone produces an identical 2048-d vector for a given image at every epoch, so computing these once and training the head on cached features is mathematically identical to training `model.fc` end-to-end, reducing a 47-minute experiment to ≈ 1 minute (18.1 s extraction for 10K train, 3.4 s for 2K validation, 0.2 s per head epoch). (ii) **GPU-resident data at reduced resolution**—raw `uint8` tensors held on the GPU (36.9 MB) with per-batch on-device resize and normalisation, and input cut from 224 to 160 px ($\approx 2\times$ cheaper). (iii) **Stratified subsampling and mixed precision**—10,000 train / 2,000 validation images, class-balanced, with AMP for runs that backpropagate through the backbone.

Pretrained vs. random initialisation. There is a big gap in accuracy: 93.3% versus 42.3% (Table 2). The from-scratch model is still improving at epoch 5, so this is not its ceiling, but the matched-budget comparison is the relevant one and it is not even close, as the from-scratch model has $2,800\times$ more trainable parameters and reaches less than half the accuracy.

A diverged run, reported as such. “Random init + head only” produced NaN losses from epoch 1 and a constant 10.0% accuracy. This is a numerical failure, not a measurement one: an untrained ResNet-152 with 151 layers of unnormalised-by-training BatchNorm produces penultimate activations of extreme magnitude, and feeding these to a linear layer under Adam at 10^{-3} overflows within a few steps. So the experiment did not even run, and the 0.1000 must not be interpreted as a result. Standardising the cached features before the probe, or lowering the learning rate, would fix it.

Head-only vs. final-block fine-tuning. Fine-tuning `layer4` plus the head (14.99M parameters) reached 92.1%—1.2 points *below* the 20K-parameter linear probe, at 31.2 s/epoch versus 0.2 s/epoch, roughly $150\times$ the cost for a worse result. This inversion is not a failure of fine-tuning in general; it is what fine-tuning looks

Table 3: Silhouette scores by layer. *Left*, Task 1.3/1.5a: “ambient” is computed on raw pooled feature vectors, the other two columns on the 2-D projections. *Right*, Task 1.5c: ambient silhouette for ResNet-152 vs. ResNet-18 on the same 2,048 images. All values are low; see Section 2.3 for why this metric is uninformative here.

Layer	ResNet-152			Ambient, depth comparison	
	Ambient	t-SNE 2-D	UMAP 2-D	ResNet-152	ResNet-18
layer1 (early)	−0.0584	−0.1258	−0.1253	−0.0584	−0.0419
layer3 (middle)	0.0000	−0.0846	−0.0827	0.0000	−0.0036
layer4 (late)	0.0114	0.0319	0.0629	0.0114	0.0222

Table 4: Task 1.5b. Cat/dog feature geometry by layer (cosine distance).

Layer	Inter-class (cat–dog)	Intra-class (cat)	Intra-class (dog)
layer1	0.0308	0.0153	0.0155
layer3	0.0487	0.0245	0.0236
layer4	0.6711	0.3997	0.4150

like when the target dataset is small. Updating 15M parameters on 10K images for five epochs perturbs well-calibrated pretrained features without re-learning better ones, and the training accuracy (0.927 vs. 0.978) confirms the model has not yet fit its data. With more epochs, more data, or a lower learning rate with warmup, we would expect fine-tuning to overtake the probe.

Which layers transfer, and the best trade-off. Early and middle layers are the most transferable, with transferability degrading towards the output. `layer1` and `layer2` compute edge, colour and texture primitives that are properties of natural images in general, so they transfer unchanged. `Layer4` encodes ImageNet-specific category abstractions. Its 2048 dimensions were shaped by the need to distinguish 1000 classes including 120 dog breeds, and is therefore most in need of adaptation but also most easily damaged by it. That the frozen `layer4` nevertheless supports 93.3% indicates the ImageNet label space is broad enough to subsume CIFAR-10’s coarse categories. The best compute/accuracy trade-off is unambiguously the frozen backbone with a linear head: highest accuracy, 0.035% of the parameters, sub-second epochs once features are cached—an optimisation available precisely *because* the backbone is frozen.

2.5 Optional Experiments

(a) t-SNE vs. UMAP. Both produce the same qualitative ordering across layers (Figure 3) and agree closely in silhouette (Table 3). UMAP scores marginally higher at `layer4` (0.063 vs. 0.032), consistent with its tendency towards more compact, better-separated clusters; t-SNE spreads points more evenly and preserves more local structure at the expense of visual tightness. Neither difference should be over-read: as Wattenberg et al. (2016) demonstrate, apparent cluster size and inter-cluster distance in these embeddings are artefacts of hyperparameters rather than properties of the data.

(b) Confusion structure. The baseline’s most frequent error is **cat predicted as dog**, occurring 112 times (Figure 5). To test whether this reflects genuine feature-space proximity we measured mean pairwise cosine distance between the two classes and each class’s internal spread at three depths (Table 4). The striking regularity is that inter-class distance is almost exactly twice intra-class distance at *every* depth. The absolute scale grows by more than an order of magnitude from `layer1` to `layer4` as representations become less redundant, but the *relative* margin does not improve with depth: deeper layers do not resolve this confusion, they magnify all distances together. Cats and dogs share pose, fur texture, facial layout and typical background, and at CIFAR-10’s native 32×32 resolution the cues that would separate them (ear shape, snout length) are largely destroyed before the network ever sees the image.

(c) ResNet-152 vs. ResNet-18. Features were extracted from the same three stages of an ImageNet-pretrained ResNet-18 over the same 2,048 images (dimensions 64, 256, 512 against 256, 1024, 2048). Silhouette scores are comparable and at `layer4` marginally favour the *shallower* network (0.0222 vs. 0.0114, Table 3). Given the metric’s demonstrated unreliability we do not conclude that ResNet-18 has better features. The defensible statement is weaker and more interesting: on this task, at this resolution, we could not detect a representation-quality advantage for Resnet 152, and Figure 6 is likewise hard to separate by eye. A plausible explanation would be that CIFAR-10’s ten coarse categories do not exercise the discriminative capacity ResNet-152’s depth provides, which was built for ImageNet’s 1000-way fine-grained problem. A linear probe on both feature could be performed for further clarification.

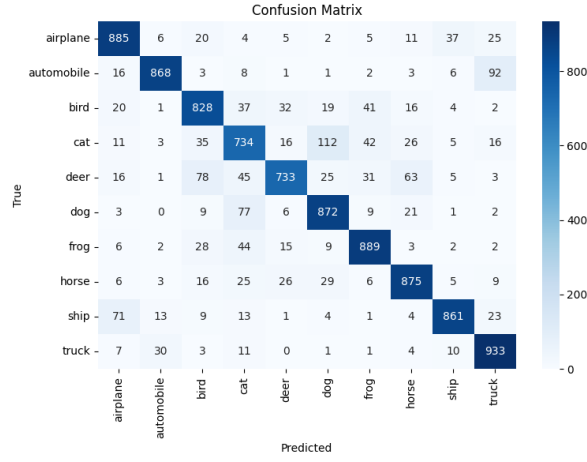


Figure 5: Task 1.5b. Confusion matrix for the frozen-backbone baseline. Errors concentrate in the animal block, with cat→dog dominant.

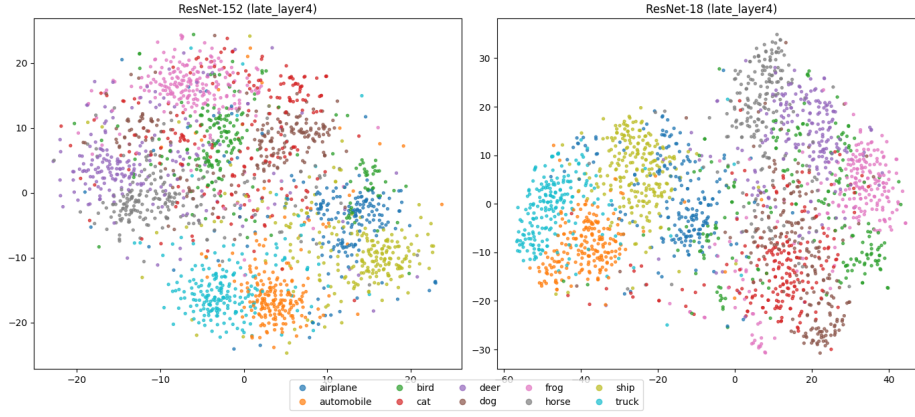


Figure 6: Task 1.5c. t-SNE of `layer4` features from ResNet-152 (left) and ResNet-18 (right) on the same images. The two are qualitatively similar.

3 Task 2: Understanding Vision Transformers

All Task 2 experiments use `google/vit-base-patch16-224` containing 12 layers, 12 heads, 768-d embeddings, 16×16 patches over a 224×224 input, giving a $14 \times 14 = 196$ patch grid plus one `[CLS]` token (`seq_len = 197`).

3.1 Pre-trained ViT for Image Classification

Three photographs (mug, shoe and wall clock taken by phone camera) were classified using the ImageNet-1k head, with the model’s own `AutoImageProcessor` handling `resize`, `rescale` and `normalisation`. All three predictions are correct (Table 5), and the confidence pattern is informative. The mug is least confident (0.615) not because the model is uncertain what the object *is*, but because ImageNet contains both “coffee mug” and “cup” as separate classes and the top two predictions sum to 0.993: the model has resolved the object and is distributing mass across a genuinely ambiguous label boundary—a calibration property raw top-1 accuracy would hide. The same holds for the clock (“wall clock” vs. “analog clock”, 0.994 combined).

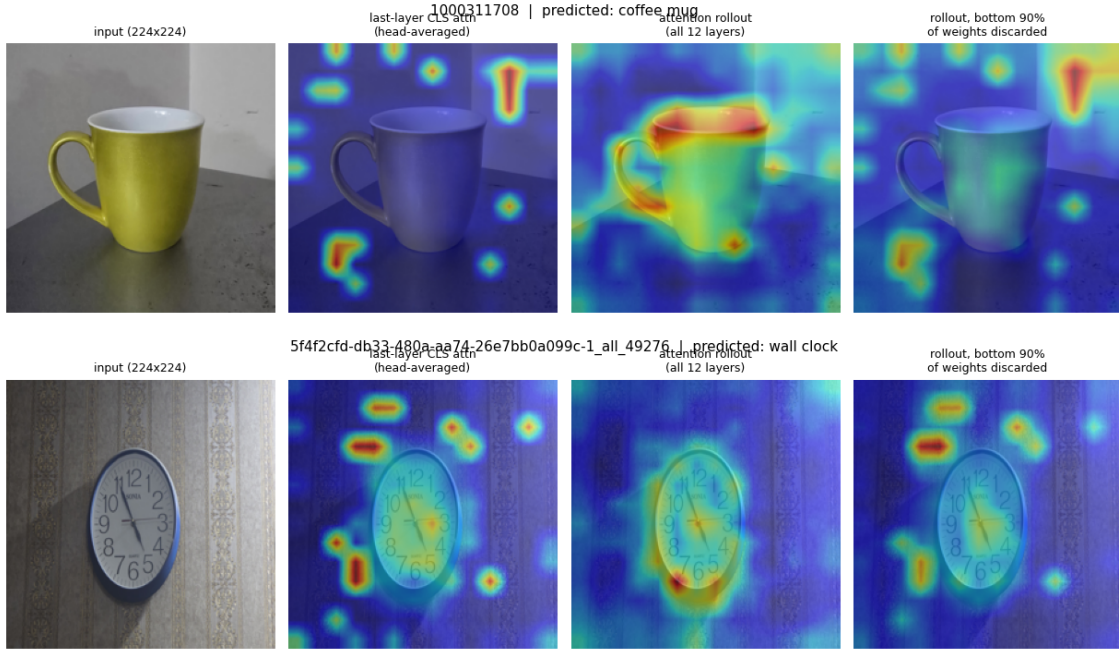
3.2 Visualizing Patch Attention

The final-layer attention tensor has shape $(1, 12, 197, 197)$. Taking row 0 (the `[CLS]` query), dropping column 0 and averaging over the 12 heads yields a 196-vector reshaped to 14×14 , bilinearly upsampled to 224×224 , min-max normalised and overlaid at 50% opacity. We also computed *attention rollout* (Abnar and Zuidema, 2020), which accounts for a layer-12 patch token already being a mixture of tokens from every earlier layer: it adds the identity to each layer’s head-averaged attention matrix (modelling the residual stream), re-normalises the rows, and multiplies across all 12 layers:

$$\tilde{A} = \prod_{l=1}^L \frac{A_l + I}{\text{rownorm}(A_l + I)}, \quad \text{attribution} = \tilde{A}_{0,1:} \quad (1)$$

Table 5: Task 2.1. Top-1 predictions and confidences.

Image	Top-1	Runner-up	$p(\text{top-1})$
Ceramic mug	coffee mug	cup (0.379)	0.615
Clock on a wall	wall clock	analog clock (0.159)	0.835
Athletic shoe	running shoe	Loafer (0.046)	0.883

**Figure 7:** Task 2.2/2.3. Raw final-layer [CLS] attention (second panel) versus attention rollout (third) and rollout with the bottom 90% of weights discarded (fourth). Raw attention lands on scattered background positions and *misses the object entirely*; rollout recovers the mug’s rim and handle, and the clock face.

The difference between raw attention and rollout. Figure 7 runs against the common assumption underlying this task. For the mug, raw final-layer [CLS] attention places its highest mass on a vertical strip in the upper right, a blob in the lower left and a small spot at centre-right, all being *background* spots and the mug itself is nearly cold. Only after rollout does the object appear, the third panel tracing the rim, body and handle of the mug with high fidelity. The same holds for the clock, where rollout concentrates on the dial while raw attention scatters across the wallpaper.

Two mechanisms explain this. First, the [CLS] representation used for classification is not built in layer 12 alone, it is accumulated through twelve residual updates, so reading only the last layer’s attention is like reading the last line of a proof and asking where the result came from; rollout’s multiplication over layers is the correct accounting. Second, the positions raw attention favours match the “attention sink” or high-norm *register token* phenomenon documented by Darcet et al. (2024): ViTs repurpose a small number of low-information patch positions as global scratchpad storage, and [CLS] attends to them heavily in late layers to retrieve aggregated global information. Those positions are largely image-independent, as can be seen from the fact that the same background blobs recur across all three test images. So the model did look at the right thing, but only the right instrument reveals it. Saying that “ViTs are interpretable because they have attention” is too superficial, as naive extraction is misleading, and a method such as rollout is required to show interpretability.

3.3 Analyzing the Attention Map and Head Specialization

Comparison with CAM / Grad-CAM. Grad-CAM weights the final convolutional feature maps by the gradient of the target class logit. Two differences follow: (i) Grad-CAM is *class-conditional*, as one can ask “what made this a cat?” versus “what made this a dog?”, whereas raw attention is class-agnostic, showing where information was gathered rather than what it supported, giving the gradient-based method a genuine advantage. (ii) Grad-CAM needs a backward pass and a target layer, and its resolution is limited by that layer’s spatial extent (typically 7×7 at 224 input), whereas ViT attention comes from a forward pass at 14×14 and exists at every layer, giving a depth-resolved view of how attribution evolves. Our results qualify the claim that transformers are “interpretable by construction”, as the mechanism is available, but not directly

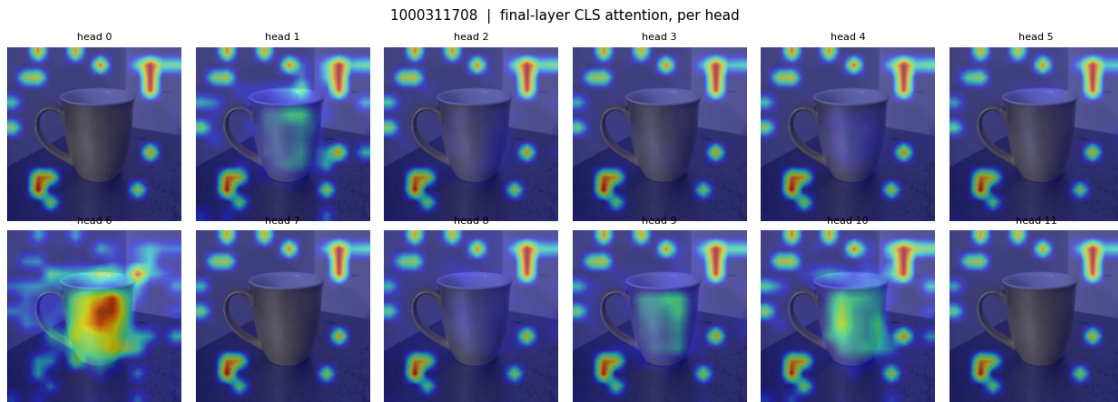


Figure 8: Task 2.3. Final-layer [CLS] attention for each of the 12 heads on the mug image. Heads 1, 6, 9 and 10 attend to the object; the remaining eight produce near-identical maps focused on the same fixed background positions.

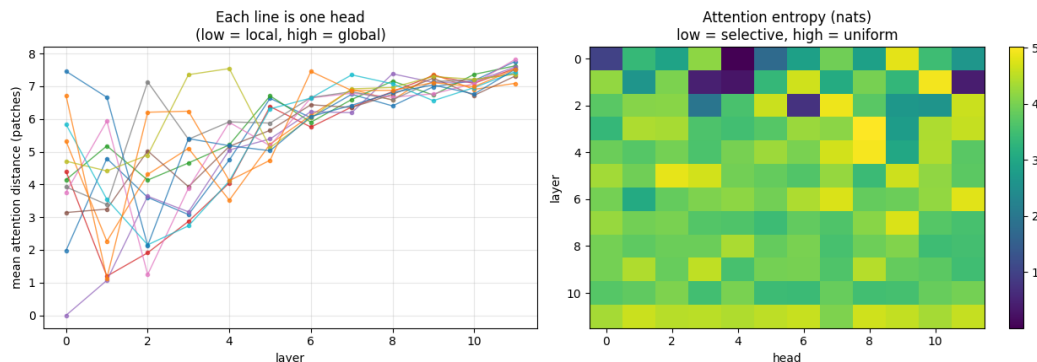


Figure 9: Task 2.3. Left: mean attention distance per head across depth (one line per head). Right: attention entropy per (layer, head). Diversity is large in layers 0–4 and collapses by layer 11.

visualisable.

Are heads specialized? Yes, but the specialization is concentrated in the *early* layers, the opposite of what one might expect. We quantified this with two per-head statistics averaged over the three images (Figure 9): attention entropy (in nats, maximum $\ln 197 = 5.28$; low means selective) and mean attention distance in patch units (maximum $\sqrt{2} \cdot 13 = 18.38$; low means local). At layer 0 the heads span the full available range (Table 6): at least one head has entropy 0.00 and attention distance 0.00, as it attends to a single token, essentially a copy operation, while another is nearly uniform at 4.82 nats. This is textbook specialization: some heads act as local edge/texture detectors, others aggregate globally. By layer 11 the spread has collapsed, entropy ranging only over [4.05, 4.66] and attention distance over [7.09, 7.81]; all twelve heads do approximately the same globally-diffuse thing, and Figure 8 confirms that eight of the twelve maps are visually interchangeable.

The diagnostic for *determining* whether specialization occurs is worth stating: compute per-head summary statistics (entropy, mean attention distance, or per-head attention-map correlation) and examine their *variance across heads* within a layer. High variance means the heads divided the labour; low variance means they are redundant. Redundancy at the top is consistent with the head-pruning literature, which finds many late-layer heads can be removed with little accuracy loss.

3.4 Patch-Masking Experiment

Patches were masked at inference time by setting them to 0.0 in *normalised* space (the per-channel dataset mean), i.e. a neutral grey. This is done as filling with black would correspond to roughly -2.1 after normalisation, an out-of-distribution input that would confound “the model lost information” with “the model was given an impossible image.” Evaluation used a logistic-regression probe fitted on *clean* [CLS] features, applied to 1,000 CIFAR-10 test images (Figure 10).

Robustness to random masking. The model is remarkably robust, as removing a quarter of the input costs 4.1 points and half still leaves 78.2% (Table 7). Two properties explain this. First, self-attention has no

Table 6: Task 2.3. Per-head statistics at three depths. The min/max range across the 12 heads is the quantity of interest.

Layer	Entropy: min / mean / max	Attn. distance: min / mean / max
0	0.00 / 2.81 / 4.82	0.00 / 4.28 / 7.46
5	3.43 / 4.12 / 4.84	4.74 / 5.69 / 6.71
11	4.05 / 4.47 / 4.66	7.09 / 7.49 / 7.81

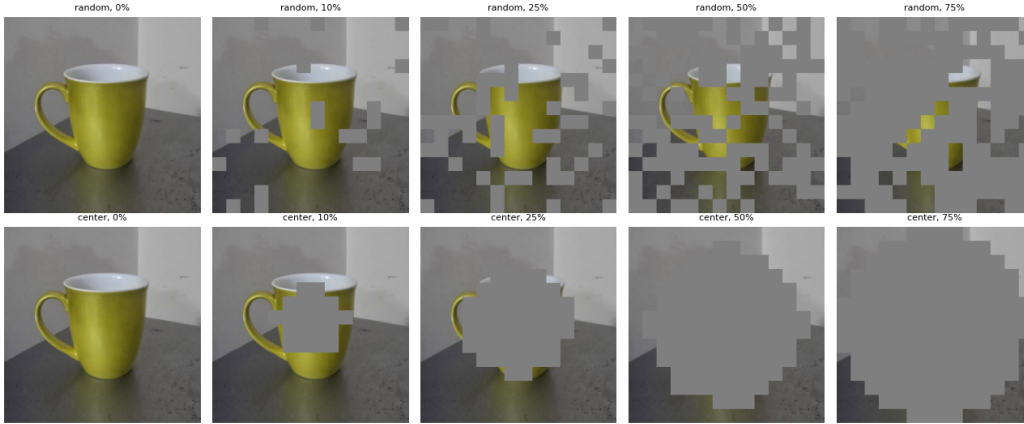


Figure 10: Task 2.4. Random (top) and centre (bottom) masking at 0, 10, 25, 50 and 75%.

architectural requirement that the input grid be complete, because each surviving token attends to every other so a hole does not break the computation the way it would break a convolution’s receptive field. Second, natural images are highly redundant: neighbouring patches are strongly correlated, so a randomly removed patch is largely predictable from its neighbours, exactly the redundancy Masked Autoencoders exploit when masking 75% of patches.

Structured masking is much worse. At 50%, centre masking costs 38.0 points against random masking’s 18.4. Random masking removes a uniform *sample*, as information is thinned everywhere but destroyed nowhere, while centre masking removes a contiguous *region*, and the correlation structure that makes random masking survivable works against contiguous occlusion: if the entire object is gone, no neighbouring context reconstructs it. In CIFAR-10 the object is essentially always centred, so centre masking is targeted removal of the signal.

Border masking improving accuracy. Masking the 10–50% of patches *furthest* from the centre raises accuracy above the unmasked baseline, peaking at 97.1% at 10% masking versus 96.6% clean. Although included as a control, it became the most interesting row in the table. The explanation is dataset-specific: CIFAR-10 images upsampled from 32×32 place the object centrally, and border patches carry background (i.e. sky, grass, road) weakly correlated with the label and acting as a distractor, so replacing it with uniform grey removes a nuisance variable. This is a genuine effect but a statement about CIFAR-10’s composition, not about ViTs; on a dataset with off-centre or context-dependent objects the sign would likely reverse. The gain (+0.5 points on 1,000 images) is within roughly one standard error, so we treat the direction as suggestive rather than established.

Stability of the ImageNet head. On the three photographs the top-1 prediction survives random masking up to 25–50% but flips at every centre masking level tested, including 10%; for the clock, centre masking at 10% drops the true-class probability from 0.835 to 0.001. This asymmetry is much sharper for single-object photographs than for the CIFAR-10 probe, as expected when the object occupies most of the frame.

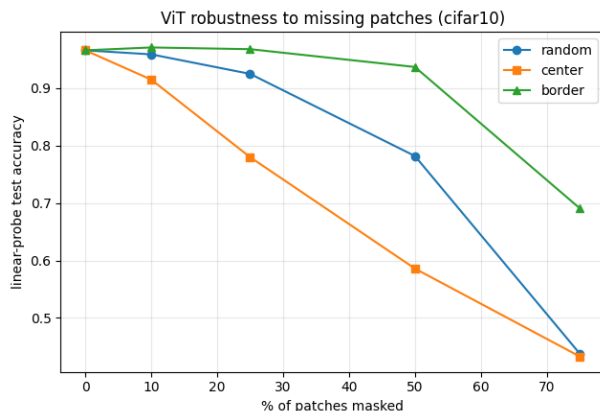
3.5 CLS Token vs. Mean-Pooled Patch Tokens

Features were extracted for 5,000 CIFAR-10 training and 2,000 test images, and logistic-regression probes fitted for four feature variants across a grid $C \in \{0.01, 0.1, 1, 10\}$.

Which is better? Mean pooling wins by 0.45 points (96.80% vs. 96.35%, Table 8), but on 2,000 test images the standard error of an accuracy near 0.966 is about 0.4 points, so this is roughly one standard error.

Table 7: Task 2.4. Linear-probe accuracy under three masking strategies, 1,000 CIFAR-10 test images.

Masked fraction	0%	10%	25%	50%	75%
Random	0.9660	0.9590	0.9250	0.7820	0.4380
Centre	0.9660	0.9150	0.7800	0.5860	0.4330
Border	0.9660	0.9710	0.9680	0.9370	0.6910

**Figure 11:** Task 2.4. Linear-probe accuracy against masked fraction for the three strategies of Table 7. Border masking sits *above* the unmasked baseline until 50%.

The honest conclusion is that the two strategies are *approximately equivalent* for this checkpoint, with a slight and not clearly significant edge to mean pooling.

Why the near-tie, and how pre-training changes it. This checkpoint was trained with supervised classification through a [CLS] head, so gradient pressure was applied specifically to make [CLS] a good summary. That it merely ties with an unweighted average of patch tokens shows how much class information is distributed across the patch tokens too. Mean pooling’s small advantage plausibly comes from variance reduction: averaging 196 tokens is a lower-variance estimate of image content than a single token routing everything through one attention bottleneck. The interaction with the pre-training objective is the general point. For objectives that never train a [CLS] token—MAE, or contrastive objectives without a class token—mean pooling is not merely competitive but necessary, since [CLS] carries no privileged meaning; conversely, for DINO-style self-distillation the objective is applied directly to [CLS], which becomes the stronger probe target. The pooling choice should follow the objective’s gradient path.

On normalisation. L2 normalisation shifts the optimal C upward by two to three orders of magnitude ($0.01 \rightarrow 10$ for [CLS]) but leaves peak accuracy unchanged—expected, since normalising to the unit sphere rescales all features so the regulariser must compensate. Feature norms are tightly concentrated ([CLS]: mean 25.57, std 0.44; mean-pool: mean 19.74, std 1.25), so norm carries almost no class information and removing it costs nothing.

4 Task 3: CLIP — Zero-Shot Classification and the Modality Gap

We used OpenAI’s CLIP ViT-B/32 (151.3M parameters, 512-d joint embedding space) and the full STL-10 test split (8,000 images, 10 classes, 800 per class). Image features were computed once and cached; text features were recomputed per prompting strategy, and stored *un-normalised* since Task 3.2 asks whether normalisation affects the modality gap.

4.1 Zero-Shot Classification on STL-10

The template helps, but more words do not. Moving from a bare class name to “a photo of a {}.” gains 1.06 points (Table 9). CLIP was trained on web alt-text, which consists of sentences rather than isolated nouns, so the template moves the text embedding into the region of embedding space the model actually saw during training—a distribution-matching effect rather than a semantic one.

The instructive result is strategy C, which *loses* 4.55 points relative to the simple template. Elaborating “a photo of a cat” into “a photo of a cat, a small domestic feline with whiskers and pointed ears” makes the prompt more informative to a human reader and less useful to CLIP: the text encoder produces a single pooled vector for the whole caption, so every additional clause pulls that vector towards the average of its

Table 8: Task 2.5. Linear-probe test accuracy (best C per variant), 2,000 CIFAR-10 test images.

Pooling	Best test accuracy	Best C
[CLS], raw	0.9635	0.01
[CLS], L2-norm	0.9630	10
Mean-pool, raw	0.9680	0.01
Mean-pool, L2-norm	0.9680	1

Table 9: Task 3.1. Zero-shot top-1 accuracy, full STL-10 test set (8,000 images), CLIP ViT-B/32.

Strategy	Example (class “cat”)	Top-1
A. Plain label	cat	96.25%
B. Photo prompt	a photo of a cat.	97.31%
C. Descriptive	...a small domestic feline with whiskers...	92.76%
D. Prompt ensemble	mean of 7 normalised templates	96.86%

constituent concepts. “Whiskers,” “pointed ears” and “feline” each have their own embedding directions, and the mixture sits further from the region occupied by actual cat *photographs* than the concise prompt does. More text dilutes rather than sharpens, a correction to the wrong intuition that richer prompts are better. The 7-template ensemble lands between the two at 96.86%; ensembling helps most when no single template is clearly best, and here “a photo of a {}” already matches STL-10 well, so averaging in “a blurry photo of a {}” adds noise.

Per-class behaviour. Accuracy ranges from 89.25% (cat) to 99.87% (ship); see Figure 12. The largest single confusion is **cat** $\rightarrow$ **deer** (51 images), not the error a human would predict, followed by car $\rightarrow$ truck (26), cat $\rightarrow$ dog (18), dog $\rightarrow$ cat (14) and deer $\rightarrow$ horse (14). The last four are semantically natural superordinate-category neighbours; cat/deer is not. Plausibly STL-10’s cat images frequently show tabby or brown cats in outdoor or grassy settings, and CLIP—trained on captions describing scenes as much as objects—weights that contextual evidence. Note that the frozen ResNet-152’s dominant error was also cat-related (cat $\rightarrow$ dog), suggesting the difficulty is a property of the category rather than of any one model.

4.2 Exploring the Modality Gap

Embeddings for 100 images (10 per class) and the 10 class text vectors were compared directly.

How separated are the modalities, and does normalisation help? The modalities are completely separated, and no, normalisation does not help. Two images of different classes are on average three times more similar to each other (cosine 0.607) than an image is to the text describing it (cosine 0.195). In Figure 13 the ten text embeddings form one tight knot well away from the image manifold, and *the text vectors are closer to each other than to any image*—including the image of the class they describe. The two modalities occupy disjoint cones of the shared 512-d sphere. Table 10 makes the normalisation question precise: the centroid ℓ_2 distance shrinks from 10.76 to 1.04, but every angular quantity is *numerically identical* to four decimal places. L2 normalisation projects both modalities onto the unit sphere, changing the radial scale of the gap without changing its angular structure; since the gap is a matter of *direction*, not magnitude, normalisation cannot close it. This is a case where a plot alone would mislead—the normalised panels look different only because t-SNE rescales its output—and only the numbers settle the question.

Why does CLIP still work? Because zero-shot classification never requires image and text embeddings to be close. The decision rule $\hat{y} = \arg \max_c \langle f_{\text{img}}(x), f_{\text{txt}}(c) \rangle / (\|f_{\text{img}}(x)\| \|f_{\text{txt}}(c)\|)$ compares *one image* against *ten texts*, so any component of the gap common to all classes—a shared offset direction separating the modalities—contributes equally to all ten scores and cancels in the $\arg \max$. What matters is the *relative* ordering within the text set, and that ordering is well calibrated even though the absolute similarities are all low. The gap arises naturally from the contrastive objective: InfoNCE only constrains matched pairs to outscore mismatched pairs, which two well-separated cones with correlated internal structure satisfy just as well as a single merged cloud. Nothing in the loss pushes the modalities together.

4.3 Bridging the Modality Gap with Orthogonal Procrustes

We sought R minimising $\|XR - Y\|_F$ subject to $R^\top R = I$, using `scipy.linalg.orthogonal_procrustes`. Crucially, R was fitted on the STL-10 *train* split (5,000 images, each paired with the text embedding of its true class) and evaluated on the *test* split, so the alignment is never fitted on the data it is

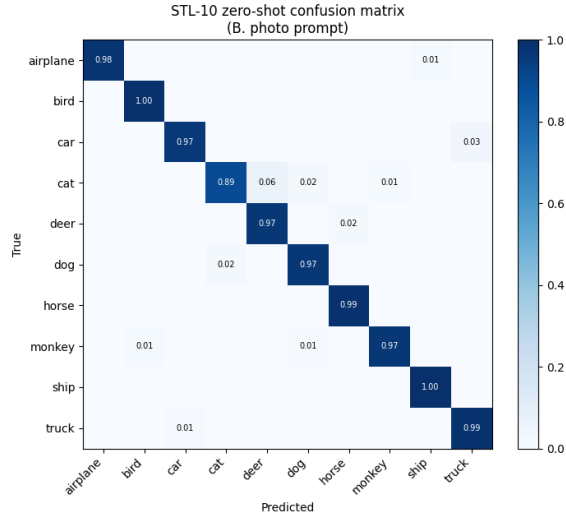


Figure 12: Task 3.1. Row-normalised confusion matrix under the best strategy.

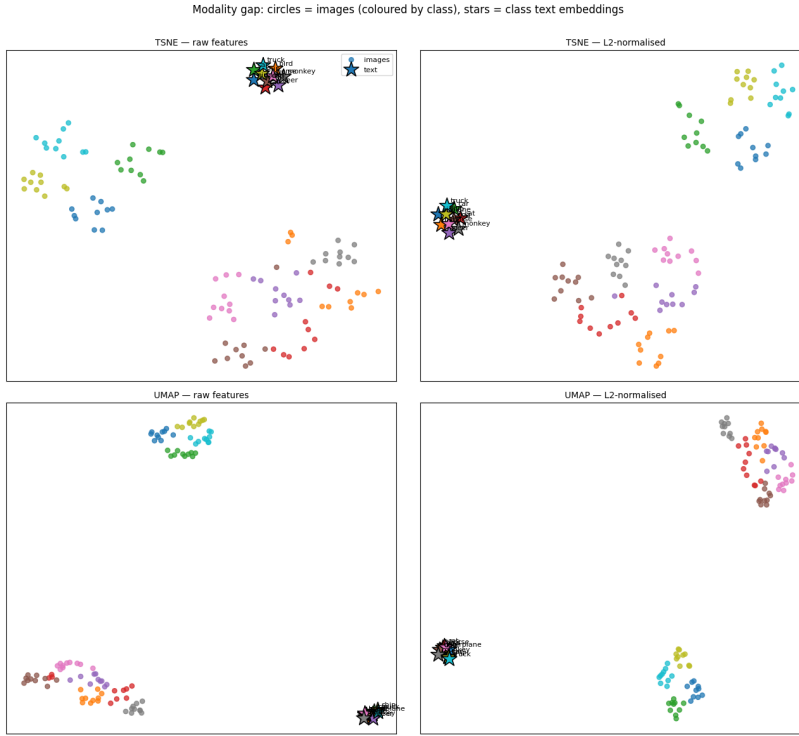


Figure 13: Task 3.2. t-SNE (top) and UMAP (bottom) of image embeddings (circles, coloured by class) and class text embeddings (stars), raw (left) and L2-normalised (right). All ten text vectors collapse into a single tight cluster far from every image, in all four panels.

scored on. Orthogonality was verified ($\|R^\top R - I\|_F = 2.6 \times 10^{-5}$) and the residual fell from 85.78 to 37.07.

The rotation closes the gap almost perfectly. Centroid cosine rises from 0.269 to 0.998 and mean cross-modal cosine from 0.195 to 0.725, now comparable to the within-image similarity of 0.607 (Table 10, Figure 14). Note that within-image cosine is *exactly* unchanged at 0.6072: an orthogonal transform preserves all inner products among the rotated set, so the image manifold’s internal geometry is untouched and only its orientation relative to the text cluster changed. This is the cleanest possible demonstration that the modality gap is a rigid relative rotation, not a difference in the structure of the two representations.

Table 10: Tasks 3.2/3.3. Modality-gap metrics with and without L2 normalisation, and after the fitted Procrustes rotation is applied to the test-set image embeddings. Only the raw centroid distance responds to normalisation.

Metric	Raw features	L2-normalised	After Procrustes
Centroid distance (ℓ_2)	10.7642	1.0416	0.1592
Centroid cosine similarity	0.2687	0.2687	0.9977
Mean within-image cosine	0.6072	0.6072	0.6072
Mean cross-modal cosine	0.1953	0.1953	0.7251

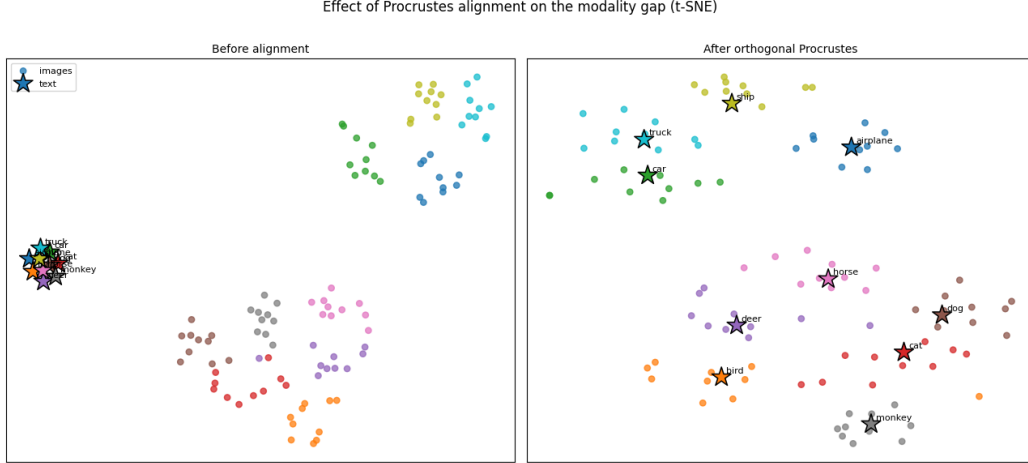


Figure 14: Task 3.3. t-SNE before (left) and after (right) the rotation. The text cluster moves into the image manifold.

However, accuracy barely moves, and the cost is high. The best strategy improves by +0.51 points, from 97.31% to 97.83% (Table 11). This near-null result is the most important finding of Task 3 and follows directly from the previous section: the gap is a shared offset that cancels in the arg max, so removing it should not help much, and it does not. However, there are two caveats. First, the aligned system is no longer zero-shot: fitting R consumed 5,000 labelled STL-10 images, and spending 5,000 labels for half a point is a poor trade. A plain logistic regression probe on the same features would very likely exceed 98%, since the Task 2 probes reached 96.8% on harder 32×32 CIFAR-10 data. Reporting 97.83% alongside genuinely zero-shot numbers without this qualification would mislead. Second, the rotation *hurt* the descriptive prompts (92.76% $\rightarrow$ 89.51%), because R was fitted using photo-prompt text embeddings as targets and does not transfer to a text distribution it was not fitted against. The alignment is specific to the prompt family used to fit it.

5 Task 4: Variational Autoencoder on MNIST

5.1 Model, Loss, and Training

The encoder is an MLP $784 \rightarrow 512 \rightarrow 256$ with ReLU activations, branching to two linear heads producing $\mu_\phi(x)$ and $\log \sigma_\phi^2(x)$. The decoder mirrors it, $d \rightarrow 256 \rightarrow 512 \rightarrow 784$, emitting *logits*. Sampling uses the reparameterisation trick $z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon$ with $\epsilon \sim \mathcal{N}(0, I)$, which keeps sampling differentiable with respect to ϕ by moving the stochasticity into a parameter-free noise variable. The loss is the negative ELBO, summed over pixels and latent dimensions and averaged over the batch:

$$-\mathcal{L}(x) = \underbrace{\sum_{i=1}^{784} \text{BCE}(\hat{x}_i, x_i)}_{\text{reconstruction}} + \underbrace{\frac{1}{2} \sum_{j=1}^d (\mu_j^2 + \sigma_j^2 - \log \sigma_j^2 - 1)}_{D_{\text{KL}}(q_\phi(z|x) \parallel \mathcal{N}(0, I))}. \quad (2)$$

Reconstruction uses `binary_cross_entropy_with_logits`, applying the sigmoid inside the loss in a numerically stable way; the closed-form KL was verified against PyTorch’s `kl_divergence` (max. absolute difference 9.5×10^{-7}). Training: Adam at 10^{-3} , batch size 128, 20 epochs on the full 60,000 MNIST training set, evaluated on the 10,000 test images, seed 0 so all latent dimensionalities share an initialisation scheme. Each run took ≈ 176 s on a T4.

However, MNIST pixels were used as continuous values in $[0, 1]$ rather than being stochastically binarised. Binary cross-entropy on continuous targets is not, strictly, a Bernoulli log-likelihood, so the ELBO values here are a valid training objective and a valid basis for internal comparison across d , but are *not* directly

Table 11: Task 3.3. Zero-shot accuracy before and after alignment, all strategies, full test set.

Strategy	A. Plain	B. Photo	C. Descriptive	D. Ensemble
Before	96.25%	97.31%	92.76%	96.86%
After Procrustes	97.12%	97.83%	89.51%	97.81%

Table 12: Task 4. Effect of latent dimensionality. All quantities per image on the 10,000-image test set after 20 epochs. The 10-NN accuracy is measured on the latent means and indicates how much class structure survives the bottleneck; the VAE never sees labels. Silhouette is reported for completeness (Task 4.2a).

d	−ELBO	Recon.	KL	KL/ d	10-NN acc.	Silhouette	Params
2	141.16	134.21	6.95	3.47	0.809	+0.121	1,068,820
5	115.48	102.92	12.56	2.51	0.935	+0.161	1,071,130
10	101.28	82.48	18.80	1.88	0.951	+0.096	1,074,980
20	100.32	80.32	20.00	1.00	0.953	+0.082	1,082,680

comparable to the ≈ 85 –90 nat figures quoted in the literature for dynamically binarised MNIST. This matches Doersch’s own treatment of pixel intensities as probabilities.

5.2 Effect of Latent Dimensionality

The total spread in test negative ELBO across $d \in \{2, \dots, 20\}$ is 40.83 nats, but excluding $d = 2$ it is only 15.16 nats, and between $d = 10$ and $d = 20$ just 0.96 nats (Figure 15). This reproduces Doersch’s observation that the VAE is insensitive to the dimensionality of z over a broad range, with degradation only at the extreme low end.

Evidence of latent-dimension collapse. The KL column tells a more precise story than the ELBO alone. Doubling d from 10 to 20 raises total KL by only 1.20 nats ($18.80 \rightarrow 20.00$), so per-dimension KL falls from 1.88 to 1.00. If all 20 dimensions carried independent information, total KL should grow roughly proportionally. Instead the model uses roughly the same latent capacity regardless of how much it is given: the surplus dimensions have their posteriors driven to match the prior ($\mu_j \approx 0$, $\sigma_j \approx 1$), contributing near-zero KL and no information. This is standard posterior collapse of unused dimensions, and it is exactly *why* the ELBO is insensitive to d in the upper range, as extra capacity is not used rather than being harmful. At $d = 2$ the situation reverses: the bottleneck binds, reconstruction cost jumps to 134.21 nats, and the model is genuinely capacity-limited.

5.3 Latent Space Visualization

At $d = 2$ the ten digits occupy coherent, contiguous wedges radiating from the origin (Figure 16), and the adjacencies are semantic rather than arbitrary: 4, 9 and 7 share the lower-left quadrant, 3, 5 and 8 the upper-centre region (confirmed independently by the decoder grid), and 0 sits diametrically opposite 1. Digits that share strokes share latent territory. The clusters touch rather than separate, which is exactly what one wants from a *generative* latent space, as the regions between clusters must decode to something plausible if prior samples are to be valid, so continuous transition is a feature of the objective rather than a failure of clustering, enforced by the KL term’s penalty on spreading posteriors apart. The aggregate posterior matches the prior well (empirical mean $[0.002, 0.056]$, standard deviation $[1.08, 1.01]$ against targets $[0, 0]$ and $[1, 1]$), though the right panel shows a mild heavy tail, with a minority of points beyond 3σ in regions the prior rarely samples—the reason VAE prior samples include occasional malformed digits.

The two metrics in Table 12 disagree in direction: 10-NN accuracy rises with d while silhouette peaks at $d = 5$ and falls. The k -NN measure is the trustworthy one, because silhouette again suffers from the distance concentration discussed in Section 2.3, and its decline at higher d reflects dimensionality rather than worse class structure. That an unsupervised model’s latent means support 95% 10-NN classification is substantive: the reconstruction objective alone recovers most of the label structure. Figure 17 decodes a grid spanning the prior’s probability mass and shows the learned manifold directly; traversing a row right to left, identity morphs continuously $0 \rightarrow 6 \rightarrow 2 \rightarrow 8 \rightarrow 1$, while higher up the path runs $0 \rightarrow 5 \rightarrow 3 \rightarrow 1$. There are no dead regions, confirming the latent space is not merely clustered but genuinely continuous.

5.4 Reconstruction Quality

Reconstructions are recognisable at every d but visibly blurred, most severely at $d = 2$, where two of the ten test digits are not constructed correctly: a 4 is reconstructed as a 9 and a 5 as a 4-like glyph (Figure 18). By $d = 10$ stroke structure is preserved and the remaining degradation is a loss of edge sharpness rather than of content.

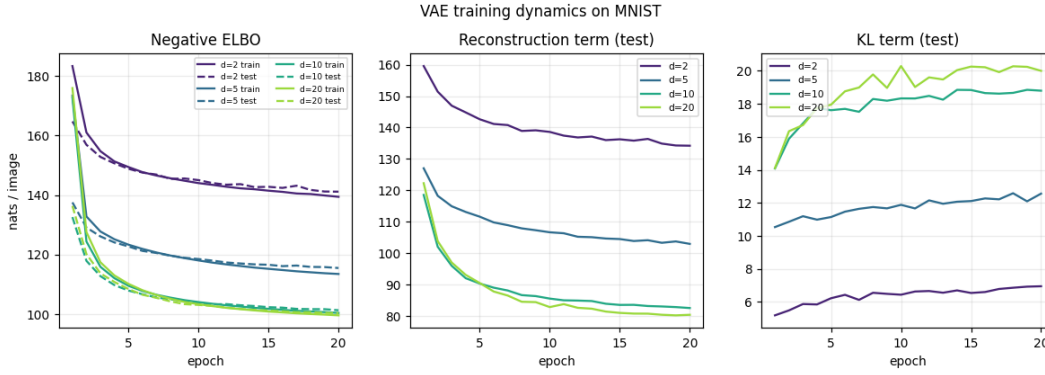


Figure 15: Task 4. Negative ELBO (left), reconstruction term (centre) and KL term (right) across 20 epochs for four latent dimensionalities. All runs converge smoothly; test and train curves stay close, indicating no overfitting.

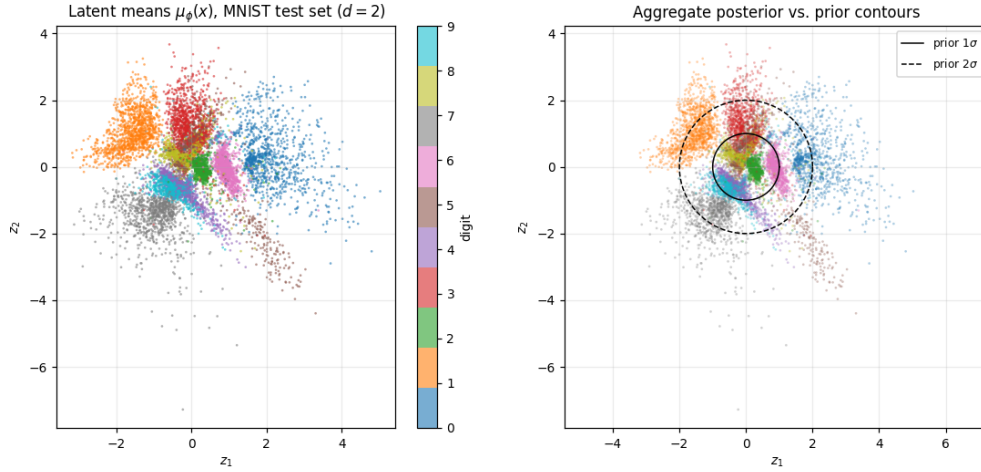


Figure 16: Task 4.2a. Latent means for the MNIST test set at $d = 2$, coloured by digit (left), and the same points against the prior’s 1σ and 2σ contours (right).

Why blurry? The Bernoulli likelihood is a per-pixel independent factorisation. Given z , the decoder cannot express correlations between neighbouring pixels, so when several plausible reconstructions are consistent with a latent code the loss-minimising output is their per-pixel *average*, and averaging sharp alternatives produces a blur. This is a property of the likelihood, not an optimisation failure, and is why VAE samples look soft next to GAN or diffusion samples.

Which digits are hardest? At $d = 10$, digit 1 costs 35.7 nats and digit 8 costs 100.2 nats—a factor of 2.8 (Figure 19), with the ordering stable across all four latent dimensionalities. The explanation is straightforward: 1 is a nearly one-dimensional family of strokes with few active pixels and little style variation, whereas 8 has two loops, many active pixels, and high variability in loop size, slant and closure. Reconstruction cost tracks the intrinsic complexity and within-class diversity of each digit, not any deficiency of the model on those classes.

5.5 Generation from the Prior, Interpolation, and Doersch Comparison

Prior samples are overwhelmingly digit-like at every d (Figure 20), confirming the KL term has shaped the aggregate posterior to overlap the prior. If it had not, samples from $\mathcal{N}(0, I)$ would land in unvisited regions and decode to noise. Diversity is good: all ten classes appear in each 64-sample grid. The failure cases are the “in-between” digits Doersch describes. At $d = 2$ several samples are ambiguous blends, with glyphs reading as a 3/8 or 4/9 hybrid, which is an expected consequence of a two-dimensional space in which ten classes must tile the plane contiguously, so any sample on a boundary between wedges decodes to a mixture. At $d = 10$ and $d = 20$ samples are sharper and more decisively one class, because extra axes let class regions separate and reduce the fraction of prior mass on boundaries. The interpolation panel shows the same property differently: walking linearly between two encoded digits passes through well-formed intermediate glyphs rather than overlaid ghosts of the endpoints, which is evidence that the space is smooth and the decoder has learned a manifold rather than a lookup table.

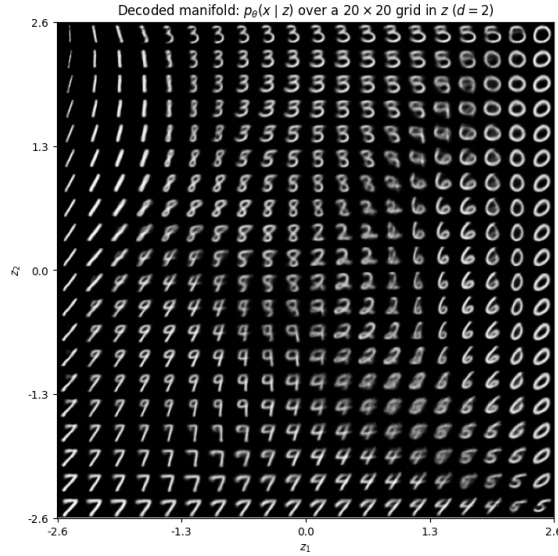


Figure 17: Task 4.2a. Decoder output over a 20×20 grid of z spaced by the inverse Gaussian CDF ($d = 2$).

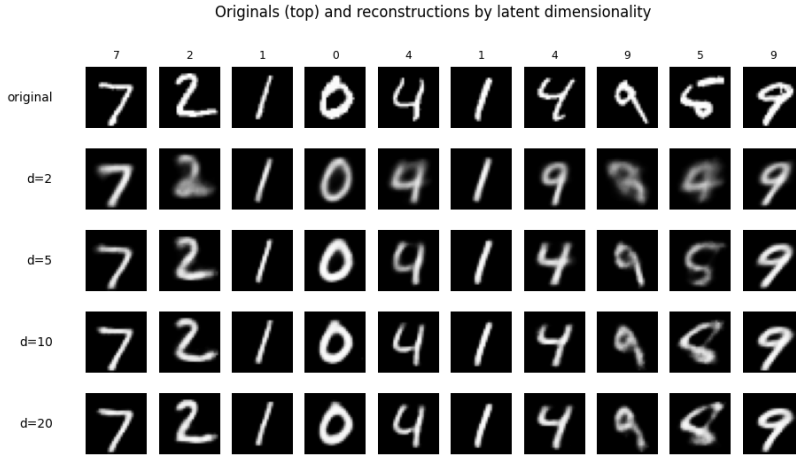


Figure 18: Task 4.2b. Ten test digits (top row) and their reconstructions at four latent dimensionalities.

Comparison with Doersch’s implementation (Task 4.3). Our results agree with [Doersch \(2016\)](#) on all four axes. *Architecture* matches in family (MLP encoder/decoder with fully connected hidden layers; ours 784–512–256 with mirrored decoder and ReLU). *Output distribution* matches: pixel intensities treated as probabilities under a per-pixel Bernoulli via sigmoid cross-entropy, implemented as `BCEWithLogits` summed over 784 pixels. *Latent-dimension insensitivity* is confirmed, as there is a 15.16 nats of spread over $d \in [5, 20]$ and 0.96 nats over $d \in [10, 20]$, with degradation only at $d = 2$ (+40.8 nats). *Sample quality* matches, with ambiguity concentrated at $d = 2$ and largely resolved by $d = 10$. The one refinement we add is a mechanism for the insensitivity: the KL/d column of Table 12 shows the model does not *use* the surplus dimensions, collapsing their posteriors to the prior. Insensitivity to d is therefore not evidence that extra dimensions are harmlessly redundant but that they are inactive, which also predicts the boundary of the effect: the ELBO becomes sensitive to d exactly when d drops below the intrinsic dimensionality the model would otherwise use.

6 Limitations and Threats to Validity

Several results rest on weaker foundations than their headline numbers suggest.

1. **Task 1.4 is not directly comparable to Task 1.1.** Task 1.1 used the full CIFAR-10 splits at 224 px; Task 1.4 a stratified 10,000/2,000 subset at 160 px. The 84.9% and 93.3% are *not* the same measurement—the difference is attributable to the smaller, class-balanced validation subset, not to any change in method. So comparison should be made within a table, not across tables.
2. **Task 1.4 configurations use different learning rates** (10^{-3} head-only, 10^{-4} when updating backbone

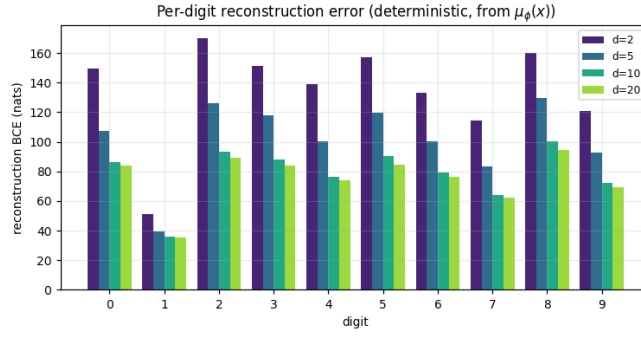


Figure 19: Task 4.2b. Per-digit reconstruction cost, decoded deterministically from $\mu_\phi(x)$.

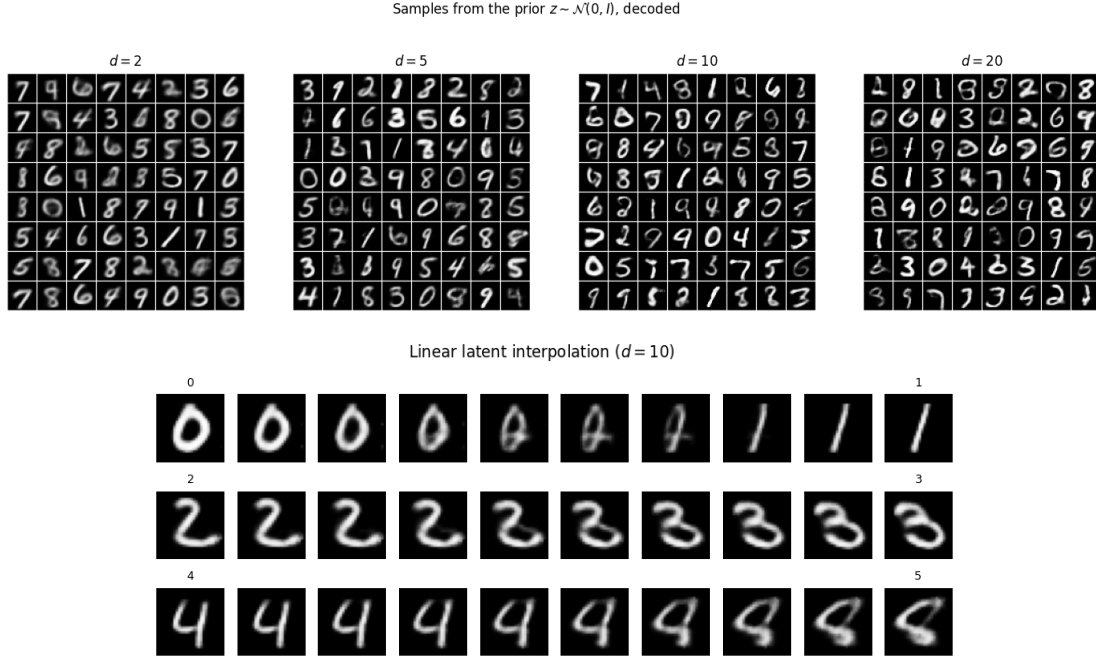


Figure 20: Task 4.2c. Top: 64 samples $z \sim \mathcal{N}(0, I)$ decoded, for each latent dimensionality. Bottom: linear interpolation between the latent means of test image pairs ($d = 10$); every intermediate point decodes to a plausible digit.

weights)—which means the head-only vs. final-block comparison confounds trainable-parameter count with optimiser setting.

3. **One Task 1.4 run diverged** (random init + head only, NaN). Its 10% accuracy is the degenerate constant predictor and carries no information about random-feature quality.
4. **Task 1.2 does not test the gradient-flow hypothesis.** With a frozen backbone no gradients traverse the ablated blocks; the experiment measures forward-pass corruption of pretrained weights.
5. **Silhouette scores are uninformative in Tasks 1.3, 1.5 and 4.2.** The metric fails under distance concentration and measures compactness rather than separability. Where it conflicts with k -NN or linear-probe accuracy, the latter should be believed.
6. **Task 2 attention analysis uses three images**—adequate for the qualitative claim that late-layer heads are redundant, far too small for the specific numerical values to be characteristic of the model.
7. **The Task 2 border-masking gain is within one standard error.** On 1,000 images, +0.5 points is not distinguishable from zero; the direction is consistent across three masking ratios, which is suggestive but not conclusive. **The Task 2.5 CLS/mean-pool gap is likewise near the noise floor** (0.45 points on 2,000 images, standard error ≈ 0.4).
8. **The Task 3 Procrustes result is not zero-shot.** It consumes 5,000 labelled examples for a +0.51-point gain and should not be listed alongside genuine zero-shot numbers without that qualification.
9. **Task 4 ELBO values are not comparable to published binarised-MNIST results**, since pixels were used as continuous values in $[0, 1]$.

7 Conclusion

Pretrained representations dominate in the small-data regime. A 20,490-parameter linear head on frozen ResNet-152 features beats a 58M-parameter network trained from scratch by more than 50 accuracy points, and beats 15M-parameter partial fine-tuning at 1/150 the cost per epoch, showing that capacity is not the binding constraint, prior is.

Interpretability tools require validation. Raw final-layer [CLS] attention in ViT does not highlight the classified object. It concentrates on fixed background positions, while attention rollout does. The claim that transformers are interpretable by construction needs the qualifier that the construction must be read correctly.

Geometry that looks fatal can be irrelevant. CLIP’s image and text embeddings occupy disjoint regions with mean cross-modal cosine 0.195, yet zero-shot accuracy is 97.31%, and closing the gap almost perfectly buys +0.51 points. A shared offset cancels in an arg max; diagnosing that analytically saved us from over-interpreting a dramatic-looking visualisation.

Negative results carry information. Descriptive prompts hurting CLIP, silhouette contradicting every other separability measure, latent dimensions collapsing to the prior, and a diverged linear probe each explained something the successful runs did not, and are reported in place rather than pruned.

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